

Chi-Square Tests, Pt 3 (REVISED)

CSU, Chico Math 351

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Introduction

χ^2 tests of independence test whether or not two random variables are independent. For example, we might want to test whether incidence rates of cancer are related to mental health or whether a politician's approval depends on the gender of the respondent. In this section we will modify the goodness-of-fit test statistic D in such a way that it can distinguish between events that are independent and events that are dependent.

Introduction

Suppose that A is some trait that has two mutually exclusive categories, A_1 and A_2 , and suppose B is some other trait with two mutually exclusive categories B_1 and B_2 . Recall that when we say that A and B are independent we can say that:

$$P(A|B) = P(A) \quad P(A) * P(B) = P(A \cap B)$$

Extending this to the case mentioned above, the definition of conditional probability indicates that:

$$P(A_i|B_j) = \frac{P(A_i \cap B_j)}{P(B_j)}$$

and rearranging the above equation gives us:

$$P(A_i|B_j) * P(B_j) = P(A_i \cap B_j)$$

Thus, we see for independent A_i and B_j :

$$P(A_i) * P(B_j) = P(A_i \cap B_j)$$

Contingency Tables

A contingency table is a table in which each observation is classified in two or more ways is called a *contingency table*. We can make a contingency table of observed counts for variables A and B with two categories each:

	B_1	B_2	Row Totals
A_1	N_{11}	N_{12}	R_1
A_2	N_{21}	N_{22}	R_2
Column Totals	C_1	C_2	n

If A and B are independent, what would be the expected counts?
We can make a contingency table of the expected counts:

	B_1	B_2
A_1	$R_1 C_1 / n$	$R_1 C_2 / n$
A_2	$R_2 C_1 / n$	$R_2 C_2 / n$

Test Statistic - General Case

We can use the χ^2 distribution to test whether two variables are independent or not. The null hypothesis is simply that the two variables are independent and the alternative hypothesis is that the two variables are dependent. The test statistic, d_2 , is as follows:

$$d_2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}$$

where r is equal to the number of rows and c is equal to the number of columns in the contingency table. d_2 can be estimated by a χ^2 distribution with $(r - 1)(c - 1)$ degrees of freedom. $\hat{E}_{ij}$ can be calculated using the following formula:

$$\hat{E}_{ij} = n * \left(\frac{R_i}{n}\right)\left(\frac{C_j}{n}\right) = \frac{R_i C_j}{n}$$

as seen on the previous slide's table for the expected counts.

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Example 1

Chase and Dunmar (1992) studied the attitudes of school-aged children in Michigan. The children were asked which of the following was most important to them: good grades, athletic ability, or popularity. Additional information about each child was also collected, and the data below shows the results for 478 children classified by sex and their response to the survey question. Test the null hypothesis that a child's answer to the survey question is independent of his or her sex, using $\alpha = 0.05$.

	Good grades	Athletic ability	Popularity
Boys	117	60	50
Girls	130	30	91

Example 1

First, we can define the table of observed counts as a matrix in R:

```
obs<-matrix(c(117,60,50,130,30,91),nrow=2,ncol=3,byrow=T)
```

To more easily calculate the expected counts, we can define the row totals, column totals, and sample size

```
r <- as.numeric(rowSums(obs,dims=1))  
c <- as.numeric(colSums(obs,dims=1))  
n <- sum(obs)  
exp <- r%*%t(c)/n # expected counts table via matrix mult.
```

Thanks to vectorization, we can calculate all parts of D at once:

```
(D <- sum((obs-exp)^2/exp) )  
[1] 21.45526
```

Example 1

For this table, $r=2$ and $c=3$, so we have $(2 - 1)(3 - 1) = 2$ degrees of freedom.

As usual, to find the probability of randomly having a difference as large or larger (under the null hypothesis that both variables are independent), we can use `pchisq()`:

`1-pchisq(21.45526, 2)  $\approx$  2.193e-05`

Our very small (smaller than $\alpha = 0.05$) p-value indicates we **REJECT** the null hypothesis that the variables are independent. The study concludes that sex has an impact on what is most important of the three options listed.

Note, though, that it does not indicate specifically how they are related, just that the two variables do not appear to be independent.

Exercise 1

A particular species of pea comes in two colors (green and yellow) and two pod shapes (full and constricted). Are green and yellow peas equally likely to have constricted pod shapes, or do they have different probabilities? Consider the following contingency table of 500 peas, and perform a Chi-Square test on the given peas dataset to determine whether or not different color peas are equally likely to have constricted pod shapes (use $\alpha = 0.05$).

	constricted	full
green	129	233
yellow	38	100

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

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Adapted from notes by Dr. Gray.